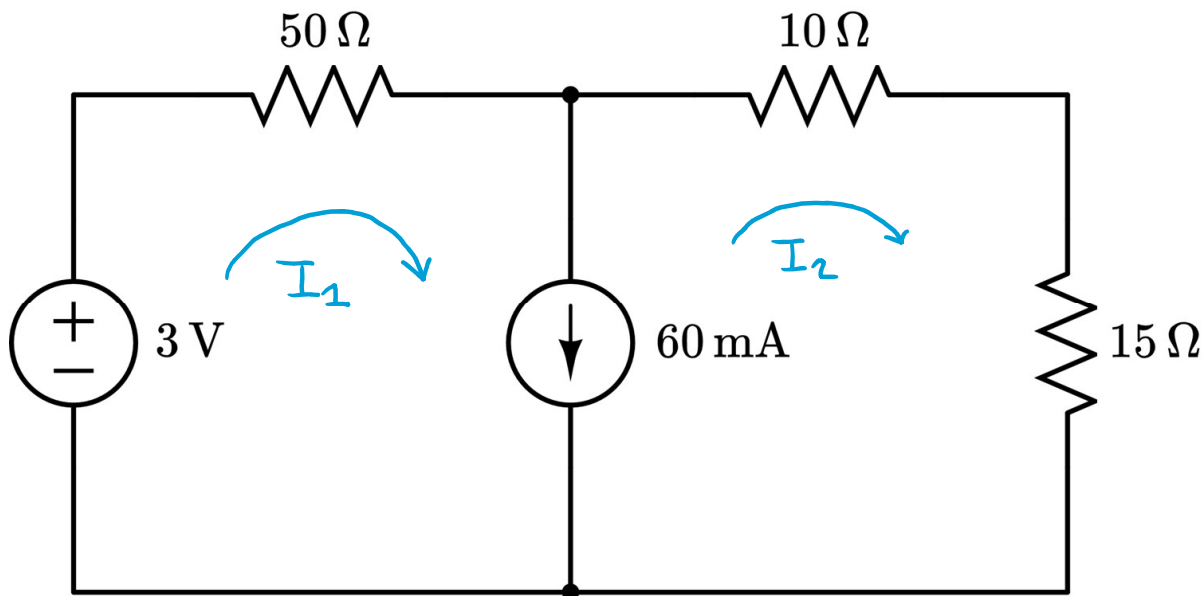


Practice Quiz Five Solution: Mesh Current Analysis (Easy)

For the circuit shown below, use mesh current analysis to define and solve for each mesh current.

Step One: Define each mesh current, identify supermesh



Supermesh: $I_2 - I_1 = 0.060$

Step Two: KVL following the Supermesh:

$$50I_1 + \underline{10I_2} + \underline{15I_2} = 3$$

$$50I_1 + 25I_2 = 3$$

Step Three: Formulate 2×2 Matrix

$$\begin{bmatrix} -1 & 1 \\ 50 & 25 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0.06 \\ 3 \end{bmatrix} \quad \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 50 & 25 \end{bmatrix}^{-1} \cdot \begin{bmatrix} 0.06 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{50} \\ \frac{2}{25} \end{bmatrix} \text{ A} = \begin{bmatrix} 20 \\ 80 \end{bmatrix} \text{ mA}$$

Self-Grading Rubric	
Clearly name and label each mesh	+2 points
Correct KVL equation following the supermesh	+2 points
Correct supermesh constraint equation	+3 points
Correct answer for i_1 and i_2	+3 points